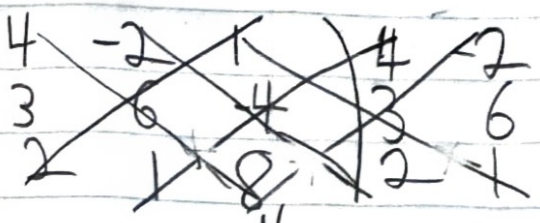
Question # 3

3a.) $A = \begin{pmatrix} 4 & -2 & 1 \\ 3 & 0 & 4 \\ 2 & 1 & 8 \end{pmatrix}$ 

$$\det(A) = [4(6)(8) + (-2)(-4)(2) + (-1)(1)(3)] - [2(6)(1) + (1)(-4)(4) + 8(3)(-2)]$$

$192 \quad 16 \quad -3$

$12 \quad -16 \quad -48$

$$\det(A) = 205 - (-52)$$

$$\boxed{\det(A) = 257}$$